Letter

Broadband quantum terahertz wave generation in laser-induced plasmas

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(Received 25 June 2025; accepted 27 January 2026; published 10 February 2026)

Terahertz (THz) radiation is widely utilized for its classical electromagnetic properties, while its quantum aspects are typically overlooked. In this work, we propose a scheme to generate quantum THz waves by employing bright squeezed vacuum states in strong laser-gas interactions. This approach significantly enhances both the bandwidth and yield of the emitted THz radiation. The resulting bunched THz photons exhibit super-Poissonian statistics, and their Wigner function corresponds to a thermal-like mixed state. The quantum nature of the generated THz waves paves the way for novel applications in sensing, communication, and beyond.

DOI: 10.1103/nssq-kk2g

Terahertz (THz) waves with low photon energies are nonionizing radiations. They have been widely used in imaging, communication, and explosive detection. In past studies, significant advances in laser technology, ultrafast optics [1,2], and nanoelectronics [3] have propelled the rapid development of THz technologies [4]. One of the major approaches for producing THz waves is the nonlinear frequency downconversion in laser-matter interactions. Various mechanisms for THz wave generation induced by lasers in gases [5–8], solid-state crystals [9–13], and liquids [14–19] have been extensively explored. These efforts have significantly expanded the accessible spectral range and enabled diverse applications [20–22]. In most current studies, both the driving laser and the generated THz wave are treated as classical electromagnetic fields, with their quantum properties typically neglected. However, nonclassical states of light at other frequencies have led to significant advances in sensing, metrology, and related fields. This highlights the strong motivation to develop broadband and efficient sources of quantum THz radiation.

Recently, the emergence of quantum light sources, particularly bright squeezed vacuum (BSV) states [23–25], has opened up new directions in strong-field physics [26,27]. Both theoretical [27,28] and experimental studies [26,29] have shown that quantum light can strongly modulate electron dynamics [30,31], thereby influencing radiation processes such as high-order harmonic generation (HHG) [32–36]. However, the potential impact of quantum light on terahertz wave generation (TWG) remains largely unexplored. In particular, it is still unknown how introducing quantum light into the driving field may alter THz emission. Investigating this question may provide valuable insights into the quantum control of low-frequency radiation and open new avenues for developing quantum THz technologies.

In this Letter, to induce quantum THz emission, we consider a strong classical driving field combined with a weak BSV field [29,37], within a one-dimensional atomic model. This configuration enables a qualitative investigation of how quantum characteristics of the driving field affect TWG and allows us to characterize the quantum features of the emitted radiation. Figure 1 illustrates the scheme in a gaseous medium. In this configuration, the strong classical field ionizes the bound electrons, while the weaker BSV field introduces quantum fluctuations that facilitate THz emissions.

In previous semiclassical frameworks [38], the electromagnetic field is treated classically while matter is treated quantum mechanically. To bridge a fully quantum framework with the semiclassical one [39], coherent states are often introduced because they offer a quantum description of classical electromagnetic fields. Building upon this foundation, the generalized P representation [40] is employed to describe a broader class of quantum optical states, including nonclassical states such as squeezed vacuum. This approach enables the investigation of quantum effects in strong-field light-matter interactions beyond the limits of classical field approximations. The density matrix of the optical field is expressed in a diagonal form, and the off-diagonal elements are neglected. It has been shown that the off-diagonal terms play minor roles in the case of the laser parameters used in this work [28].

Under the limiting case $\varepsilon^{(1)} \to 0$, a previous study [27] has derived the following expression for the emission spectrum:

$$S(\omega) = \frac{\omega^4}{6\pi^2 c^3 \epsilon_0} \int d^2\mathcal{E}_\alpha ||\phi_\alpha||^2 Q(\mathcal{E}_\alpha)|\boldsymbol{d}_\alpha(\omega)|^2. \quad (1)$$

Here, the normalization condition $||\phi_\alpha||^2 = 1$ ensures that the time-evolved electronic wave function $\phi_\alpha(t)$ remains

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normalized. The function $Q(\mathcal{E}_\alpha)$ represents the Husimi Q distribution [41] of the initial optical state $\hat{\rho}_0$, with $\mathcal{E}_\alpha$ representing the complex amplitude associated with the coherent state $|\alpha\rangle$. The quantity $\boldsymbol{d}_\alpha(\omega)$ corresponds to the dipole radiation response at frequency ω, obtained from the classical trajectory initialized with the field amplitude $\mathcal{E}_\alpha$. This formalism can be naturally extended to describe current-based radiation by replacing the dipole amplitude with the frequency-domain current response $\boldsymbol{j}_\alpha(\omega)$. The resulting spectral power density reads

$$S(\omega) = \frac{\omega^2}{6\pi^2 c^3 \epsilon_0} \int d^2\mathcal{E}_\alpha Q(\mathcal{E}_\alpha)|\boldsymbol{j}_\alpha(\omega)|^2. \tag{2}$$

In this formulation, $\boldsymbol{j}_\alpha(\omega)$ denotes the Fourier component of the current response at frequency ω. The normalization factor $||\phi_\alpha||^2$ is omitted here since it is identically equal to one by construction.

In the theoretical simulation, the total optical field can be described by the product state: $|\chi\rangle = |\alpha_0\rangle_{\rm L} \otimes \hat{S}(\zeta)\,|0\rangle_{\rm 2L}$, where $|\alpha_0\rangle_{\rm L}$ is a coherent state at frequency $\omega_{\rm L}$, and $\hat{S}(\zeta)\,|0\rangle_{\rm 2L}$ is a squeezed vacuum state centered at frequency $2\omega_{\rm L}$.

The THz yield can be evaluated as

$$\begin{aligned} S(\omega) &= \frac{\omega^2}{6\pi^2 c^3 \epsilon_0} \iint d^2\mathcal{E}_\alpha d^2\mathcal{E}_\beta Q_{\rm CS}(\mathcal{E}_\alpha) Q_{\rm BSV}(\mathcal{E}_\beta)|\boldsymbol{j}_{\alpha,\beta}(\omega)|^2 \\ &\approx \frac{\omega^2}{6\pi^2 c^3 \epsilon_0} \iint d^2\mathcal{E}_\alpha d^2\mathcal{E}_\beta \delta^{(2)}(\mathcal{E}_\alpha - \mathcal{E}_{\alpha_0}) Q_{\rm BSV}(\mathcal{E}_\beta) \\ &\quad \times |\boldsymbol{j}_{\alpha,\beta}(\omega)|^2 \\ &\approx \frac{\omega^2}{6\pi^2 c^3 \epsilon_0} \int d^2\mathcal{E}_\beta Q_{\rm BSV}(\mathcal{E}_\beta)|\boldsymbol{j}_{\alpha_0,\beta}(\omega)|^2. \end{aligned} \tag{3}$$

Here, $Q_{\rm CS}(\mathcal{E}_\alpha)$ and $Q_{\rm BSV}(\mathcal{E}_\beta)$ denote the Husimi Q distributions of the coherent and BSV components, respectively. The current response $\boldsymbol{j}_{\alpha_0,\beta}(\omega)$ is computed from a trajectory driven by the composite state $|\alpha_0\rangle_{\rm L} \otimes |\beta\rangle_{\rm 2L}$, with constituent frequencies $\omega_{\rm L}$ and $2\omega_{\rm L}$.

To reduce computational cost, particularly in higher-dimensional parameter spaces, the integral in Eq. (2) can be approximated using a Monte Carlo sampling scheme. Owing to the positivity of the Husimi distribution $Q(\alpha)$, we draw a set of samples $\{\beta_n\}$ distributed according to $Q_{\rm BSV}(\mathcal{E}_\beta)$, and estimate the spectral yield via

$$S(\omega) = \frac{\omega^2}{6\pi^2 c^3 \epsilon_0} \frac{1}{N} \sum_{n=1}^{N} |\boldsymbol{j}_{\alpha_0,\beta_n}(\omega)|^2. \tag{4}$$

For each sampled state β_n, the current response $\boldsymbol{j}_{\alpha_0,\beta_n}(\omega)$ is calculated under the influence of a combined optical field, modeled as the product state $|\chi_n\rangle = |\alpha_0\rangle_{\rm L} \otimes |\beta_n\rangle_{2L}$. The corresponding electric field acting on the electron is given by

$$\begin{aligned} \vec{E}_{\alpha_0,\beta_n}(t) = {}& 2g_{0,1}\sqrt{\omega_L}|\alpha_0|\sin(\omega_L t - \theta_{\alpha_0}) \\ &+ 2g_{0,2}\sqrt{2\omega_L}|\beta_n|\sin(2\omega_L t - \theta_{\beta_n}), \end{aligned} \tag{5}$$

where θ_{α_0} and θ_{β_n} are the phases of α_0 and β_n, respectively, and g_0 is the coupling strength which can be calculated by $g_{0,1} = \frac{E_{0,1}}{2\sqrt{\omega_L}|\alpha_0|}$, and $g_{0,2} = \frac{E_{0,2}}{2\sqrt{2\omega_L}|\beta_n|}$, where E_0 is the field strength.

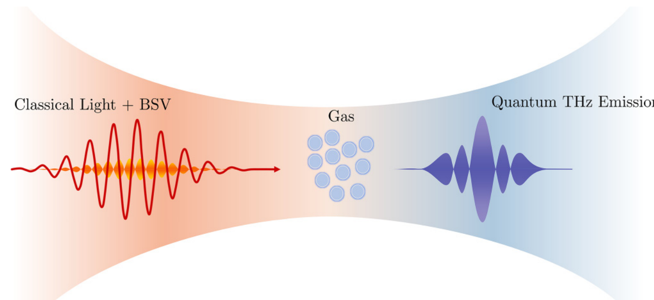

FIG. 1. The interaction of a two-color field and a gas to generate THz waves. The composite field comprises classical light and BSV with wavelength ratio 2:1. The THz emission carries quantum properties.

To determine the time-dependent current, we solve the semiclassical time-dependent Schrödinger equation (TDSE):

$$\begin{aligned} i\frac{\partial}{\partial t}\,|\psi_{\alpha_0,\beta_n}(t)\rangle &= \hat{H}_{\alpha_0,\beta_n}\,|\psi_{\alpha_0,\beta_n}(t)\rangle\,, \\ \hat{H}_{\alpha_0,\beta_n} &= \frac{p^2}{2} + \hat{V} + \hat{x}\cdot\vec{E}_{\alpha_0,\beta_n}(t), \end{aligned} \tag{6}$$

where $|\psi_{\alpha_0,\beta_n}(0)\rangle$ is the ground state of the unperturbed atomic system.

The current in the frequency domain is then obtained from the Fourier transform of the expectation value of the momentum operator:

$$\begin{aligned} \boldsymbol{j}_{\alpha_0,\beta_n}(\omega) &= \mathcal{F}\{\boldsymbol{j}_{\alpha_0,\beta_n}(t)\}, \\ \boldsymbol{j}_{\alpha_0,\beta_n}(t) &= \langle\psi_{\alpha_0,\beta_n}(t)|\hat{p}|\psi_{\alpha_0,\beta_n}(t)\rangle\,. \end{aligned} \tag{7}$$

In actual calculations, both the classical 800 nm field and the 400 nm BSV field are modeled with Gaussian envelopes, each having a full width at half maximum (FWHM) of approximately 60 fs. For simulations at other central wavelengths, the FWHM is scaled proportionally with wavelength to maintain a consistent number of optical cycles.

In the classical picture, the TWG mainly originates from the broken symmetry of the field or the system in real space [5]. Figure 2(a) shows a representative THz spectrum, which is on the noise level if only the 800 nm classical field is applied since the symmetry is preserved. In contrast, for the case of the two-color laser by classical 800 nm field and classical 400 nm field with phase difference $\pi/2$, the THz yield can be enhanced by two orders. The BSV light is with zero mean electric field and large photon-number fluctuations. The replacement of a classical 400 nm field by a 400 nm BSV field with the same phase difference can give stronger THz radiation. For the origin of the THz field in this combined field, the classical IR field ionizes the gas to generate quasifree electrons, and then the IR + BSV fields drive the quasi-free electrons to form THz oscillations. The BSV field can be expanded by coherent states, and the electron motion can be decomposed into a series of trajectories in the synthesized nonsymmetric field with different amplitudes. The final THz emission is a mixture of the contributions from each trajectory. The mechanism can also be understood qualitatively by the four-wave-mixing model [42] in which the instantaneous third-order polarization will lead to a rectified term

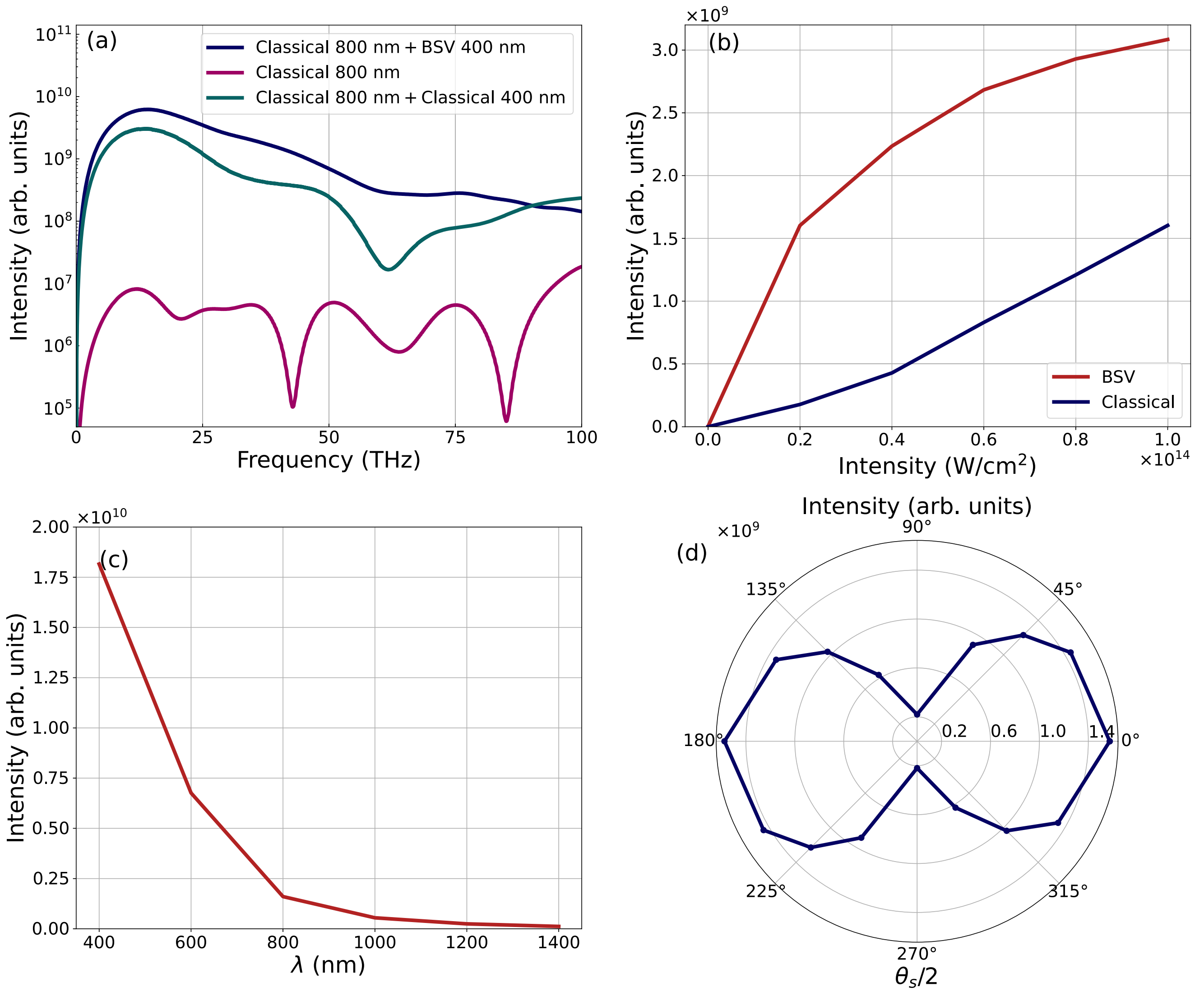

FIG. 2. (a) Typical THz spectrum calculated under an 800 nm classical field and a 400 nm BSV field. The respective peak intensities are 1×10^{14} W/cm^2 (classical) and 2×10^{13} W/cm^2 (BSV), respectively. For comparison, the THz radiations generated in the monochromatic 800 nm classical field and the two-color field by classical 800 nm and classical 400 nm lasers with a phase difference of $\pi/2$ are also illustrated. (b) Dependence of THz yield on the intensity of the classical and BSV fields. The intensity of one field is varied while the other is kept constant: 2×10^{13} W/cm^2 for the BSV field and 1×10^{14} W/cm^2 for the classical field. (c) Dependence of THz yield on the wavelength of the two-color field. Both field intensities are fixed at the same values, i.e., 2×10^{13} W/cm^2 for the BSV field and 1×10^{14} W/cm^2 for the classical field. (d) THz yield as a function of the squeezing angle θ_s of the BSV field. The variation reflects the phase-sensitive interference between the quantum and classical driving fields.

responsible for the THz emission. To quantify the THz yield, we define an average intensity over the THz frequency range as $\bar{S} = \frac{1}{\omega_c}\int_0^{\omega_c} d\omega S(\omega)$. Next, we investigate the scaling of the THz yield with respect to the driving intensities. Specifically, we analyze the dependence of $\bar{S}$ on both classical and BSV field intensities, as shown in Fig. 2(b). The THz yield increases monotonically with both classical and quantum field intensities. Within the examined range, it exhibits an approximately quadratic dependence on the classical intensity. In contrast, the response to the BSV field displays a clear saturation behavior at higher intensities. This saturation suggests that quantum nonlinear effects emerge, possibly due to the limited phase space or the depletion of excitation channels in the quantum field.

The wavelength dependence of the THz yield under the same calculation method is shown in Fig. 2(c). Throughout this study, the wavelength ratio between the classical and BSV fields is fixed at 2:1. The intensities are constant at 1×10^{14} W/cm^2 (classical) and 2×10^{13} W/cm^2 (BSV). Our results indicate that shorter fundamental wavelengths lead to significantly higher THz yields.

We further examine the influence of the squeezing angle θ_s on TWG, as presented in Fig. 2(d). This squeezing angle determines the quantum field's phase-space orientation, thereby controlling the relative phase between the BSV and classical driving fields. In our Monte Carlo model, this manifests as a rotation of the BSV phase-space samples $\{\beta_n\}$.

When the squeezing angle $\theta_s = 0$, the sampling phases θ_{β_n} cluster around $\pi/2$ and $3\pi/2$, leading to strong constructive interference with the classical field, and hence a high THz yield. Conversely, when $\theta_s = \pi$ (i.e., $\theta_s/2 = 90^\circ$), the sampling phases rotate to 0 and π, which corresponds to destructive interference and results in suppressed TWG. This dependence reflects the sensitivity of the nonlinear electron dynamics to the relative optical phase, particularly in the quantum-enhanced regime enabled by BSV illumination.

The classical electromagnetic field corresponds most closely to the coherent state. To investigate the nonclassical properties of the THz emission, we study its quantum statistical characteristics to find its difference from the coherent state, like the photon number distribution and the Wigner function. The photon number distribution can be extracted from the Husimi Q function of the BSV light. Specifically, the probability of detecting N photons at frequency ω is given by [28]

$$P_{\boldsymbol{k}}(N) = \int d\beta Q(\beta) e^{-\left|\gamma_{\boldsymbol{k}}^{\alpha_0,\beta}\right|^2} \frac{\left|\gamma_{\boldsymbol{k}}^{\alpha_0,\beta}\right|^{2N}}{N!}, \tag{8}$$

where $\gamma_{\boldsymbol{k}}^{\alpha_0,\beta}$ denotes the mode amplitude associated with the optical mode $\boldsymbol{k}$, determined by the current response of the system. For numerical evaluation, this distribution can be approximated via Monte Carlo sampling over the Q distribution:

$$P_{\boldsymbol{k}}(N) \approx \frac{1}{N} \sum_{n=1}^{N} e^{-\left|\gamma_{\boldsymbol{k}}^{\alpha_0,\beta_n}\right|^2} \frac{\left|\gamma_{\boldsymbol{k}}^{\alpha_0,\beta_n}\right|^{2N}}{N!}. \tag{9}$$

Each summation term in Eq. (9) corresponds to a Poisson distribution associated with a particular phase-space sample β_n, representing a coherent component of the light field. However, it is important to note that the calculated current amplitudes $|\boldsymbol{j}_{\alpha_0,\beta}(\omega)|$ are extremely small due to the simulation representing interaction with a single atom or molecule. The average photon number is too small to reflect the distribution. In realistic experimental scenarios, the laser interacts with a macroscopic ensemble of emitters (e.g., gas atoms), leading to a collective enhancement of the emitted harmonics. To account for this macroscopic scaling, we introduce a coefficient $\boldsymbol{s} = 3 \times 10^6$, such that [43,44] $\gamma_{\boldsymbol{k}}^{\alpha_0,\beta_n} = -ig_0\boldsymbol{s} \cdot \frac{j^*_{\alpha_0,\beta}(\omega_k)}{\sqrt{\omega_k}}$, allowing us to match the estimated photon numbers to experimentally accessible levels. However, $\boldsymbol{s}$ will change the normalized photon number distribution. When $\boldsymbol{s}$ increases, it corresponds to higher intensities, and the distribution will shift to higher photon numbers. Then we calculate the distribution of the photons across the THz range by the equation

$$P_{\mathrm{THz}}(N) = \frac{\sum_{\boldsymbol{k}} P_{\boldsymbol{k}}(N)}{\mathrm{N}_{\boldsymbol{k}}} = \frac{\int_0^{\omega_c} d\omega \omega^2 P_\omega(N)}{\int_0^{\omega_c} d\omega \omega^2}. \tag{10}$$

Here, $P_{\boldsymbol{k}}(N)$ stands for photon distribution for mode $\boldsymbol{k}$, and has the relation $P_{\omega_k}(N) = P_{\boldsymbol{k}}(N)$. $\mathrm{N}_{\boldsymbol{k}}$ denotes the total number of modes considered in the $\boldsymbol{k}$ space for the calculation. Because $\sum_{\boldsymbol{k}} \approx \frac{V}{8c^3\pi^3}\int d\Omega d\omega\omega^2$, we only consider one polarization σ for a one-dimensional system. For clarity, the maximum value of the distribution is scaled to one.

Figure 3 presents the resulting photon number distribution. The distribution is THz-frequency dependent. The overall shapes are similar, but shift to low photon numbers as the frequency increases. To capture the overall characteristics of the TWG, we also sum the photon distributions over the frequency range 1–100 THz, providing a broader statistical characterization of the emitted radiation. As shown in Fig. 3(a), compared with the Poisson distribution of the coherent state, the photon number distribution in the THz range exhibits clear Bose-Einstein distribution of thermal light. This suggests that the THz emission exhibits quantum characteristics, consistent with the results in Fig. 2(a), which indicate that the signal originates primarily from the BSV field and thus retains its quantum properties. For comparison, we also compute the photon number distribution of the third harmonic H3 in Fig. 3(b). As we discussed above, the mechanism of HHG is different from the THz wave generation. It displays a Poisson-like distribution characteristic of a coherent state, agreeing with the experimental measurement in Ref. [29]. This shows that the odd harmonic signal primarily reflects the classical driving field, yet it is modified by the BSV field, which imparts its quantum character.

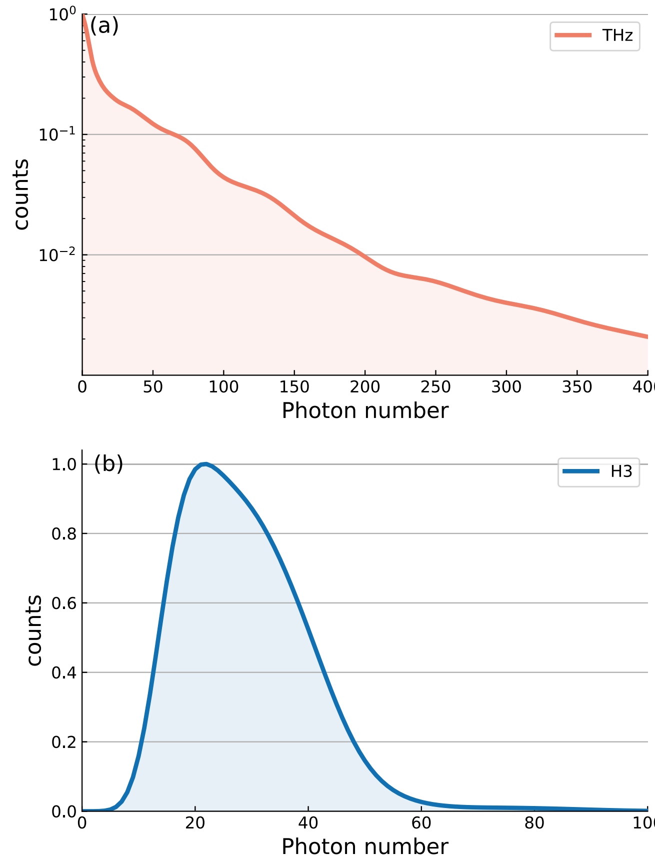

FIG. 3. The photon distribution of the (a) THz range 1–100 THz and (b) H3, respectively.

To further investigate the quantum characteristics of the generated radiation, we compute the Wigner function for the optical field at a specific mode $\boldsymbol{k}$, defined as $W_{\boldsymbol{k}}(\alpha) = \frac{1}{\pi^2}\int d^2\lambda \mathrm{Tr}[\hat{\rho}_{\boldsymbol{k}}^R \hat{D}_{\boldsymbol{k}}(\lambda)] e^{\alpha\lambda^* - \alpha^*\lambda}$ [45]. Here, $\hat{\rho}_{\boldsymbol{k}}^R$ denotes the reduced density matrix of the optical field at mode $\boldsymbol{k}$, and $\hat{D}_{\boldsymbol{k}}(\lambda)$ is the displacement operator.

To construct $\hat{\rho}_{\boldsymbol{k}}^R$, we approximate it as a statistical mixture of coherent states [27]: $\hat{\rho}_{\boldsymbol{k}}^R \approx \int d\beta Q(\beta) |\gamma_{\boldsymbol{k}}^{\alpha_0,\beta_n}\rangle\langle\gamma_{\boldsymbol{k}}^{\alpha_0,\beta_n}|$, or, numerically, as a finite

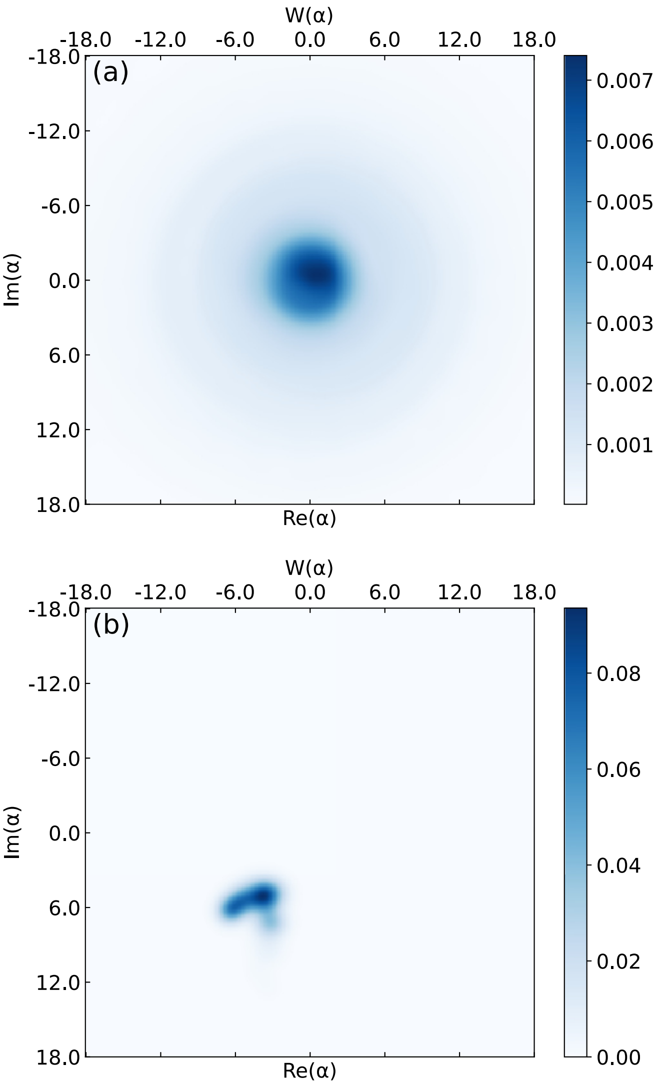

FIG. 4. The Wigner function of the optical field (a) THz range 1–100 THz and (b) H3. The phase-space ranges are kept identical to facilitate direct comparison of the distributions.

ensemble: $\hat{\rho}_{\boldsymbol{k}}^R \approx \frac{1}{N}\sum_{n=1}^{N} |\gamma_{\boldsymbol{k}}^{\alpha_0,\beta_n}\rangle\,\langle\gamma_{\boldsymbol{k}}^{\alpha_0,\beta_n}|$. This form corresponds to a mixed coherent state at mode $\boldsymbol{k}$. In fact, the photon distribution can also be examined as Eq. (9) by $P_{\boldsymbol{k}}(N) = \mathrm{Tr}[\hat{\rho}_{\boldsymbol{k}}^R\,|N\rangle\,\langle N|_{\boldsymbol{k}}]$. For numerical implementation, we normalize the density matrix to satisfy $\mathrm{Tr}[\hat{\rho}_{\boldsymbol{k}}^R] = 1$.

Once the reduced density matrix $\hat{\rho}_{\boldsymbol{k}}^R$ is obtained, the Wigner distribution in the THz range is computed by averaging over the relevant frequency band $W_{\mathrm{THz}}(\alpha) = \int_0^{\omega_c} d\omega\omega^2 W_\omega(\alpha)/\int_0^{\omega_c} d\omega\omega^2$, which is similar to a photon distribution situation. For comparison, we also examine the emission spectrum of odd-order harmonics. From Fig. 4(a), the Wigner function of the THz range is broader and centered at the origin, indicating that the THz radiation corresponds to an approximately symmetric, thermal-like mixed state. Our calculation suggests that the quantum nature of the generated THz radiation is not characterized by squeezed, narrow distributions, but rather by a thermal-like profile without evident squeezing. This feature is also in accordance with the photon number distribution in Fig. 3(a). In contrast, in Fig. 4(b), the Wigner function of H3 exhibits a localized, coherent-state-like distribution with quantum fluctuations coming from the BSV driving field. This field alters the radial and angular distribution of the H3 obviously in the phase space.

In summary, we proposed an efficient scheme to produce quantum THz emissions by introducing BSV light in the classical field. We thus generate quantum THz waves in laser-gas interactions. A Monte Carlo sampling approach is employed to calculate and analyze the properties of THz radiation. It reveals that the THz yield sensitively depends on the intensity, wavelength, and the squeeze angle of the driving fields. Furthermore, its quantum features are clearly identified by the photon number distribution and Wigner function. It will open a door for generating and utilizing the quantum properties of THz waves in future studies.

This work is supported by the National Natural Science Foundation of China (NSFC) (Grant No. 12274421), the Hubei Provincial Natural Science Foundation of China (Grant No. 2024AFA029), and the Guangdong Provincial Key Laboratory of Materials and Technologies for Energy Conversion (Grant No. MATEC2024KF001). M.F.C. acknowledges financial support from the National Key Research and Development Program of China (Grant No. 2023YFA1407100), the Guangdong Province Science and Technology Major Project (Future functional materials under extreme conditions-2021B0301030005), and the Guangdong Natural Science Foundation (General Program Project No. 2023A1515010871).

Data availability. The data that support the findings of this article are not publicly available. The data are available from the authors upon reasonable request.

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